Faculty of Science and Technology

Programme Structure & Syllabus

For

First Year

Bachelor of Technology (Computer Science and Engineering)

Programme Code: BTECCSE

Pattern 2025

With Effect from Academic Year 2025 – 2026

	Authority	Date
Proposed by	Board of Studies in Department of Engineering Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS **Dean of Faculty** **Director, IQAC**

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Engineering	
To create an intellectual and academically rich environment for careers in Computing Education and Research to fulfill global needs.	
Mission of the Department of Computer Engineering	
M1	To impart modern computing education through novel methods and research.
M2	To prepare the students for life-long learning in pursuit of excellence.
M3	To create technically competent human in socio-economic domain.
M4	To promote aspiring ethically conscious engineers demonstrating sustainable entrepreneurship and professional maturity in social context.

Values of the Department of Computer Engineering	
Teamwork.	Co-operation.
Understanding.	Integrity.

Programme Educational Objectives (PEOs)

PEO No.	Statement	
PEO1	**Preparation**	To prepare the students to evolve into a professional and committed data science technology workforce, by providing them with a global educational platform with innovative practices.
PEO2	**Core competence**	To provide a sufficient mathematical, statistical and computing theory knowledge base, leading to practical and long lasting data science solutions.
PEO3	**Breadth**	To ingrain problem solving skills in data science and sound engineering principles in students with the help of modern data science tools and techniques.
PEO4	**Professionalism**	To impart in the students professional and ethical practices based on standard guidelines as applied to the field of artificial intelligence to acquire the right attitude and aptitude for the benefit of society.
PEO5	**Learning Environment**	To guide students for higher education and lead them to avenues of entrepreneurship in areas of artificial intelligence and data science.

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3	PEO4
M1	1		1	
M2	1	3	3	2
M3	1			3
M4	2	2		

Affinity Level: 1- Slight, 2- Moderate,3-Substantial

Programme Outcomes (POs)

PO No.	Statement	
	Graduate Attributes	**Programme Outcome**
PO1	**GA 1:** Engineering Knowledge	Apply the knowledge of engineering, data science fundamentals and artificial intelligence to solve complex engineering problems.
PO2	**GA 2:** Problem Analysis	Identify, formulate and analyze complex engineering problems to create solutions using the first principles of mathematics, statistics, and data science.
PO3	**GA 3:** Design/Development of solution	Design solutions for system components that meet the specified needs for the benefit of society.
PO4	**GA 4:** Conduct investigation of Complex Problems	Apply research methods including design of experiments, statistical analysis and business interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**GA 5:** Modern Tool Usage	Apply appropriate data mining techniques and statistical tools to model complex engineering activities.
PO6	**GA 6:** The Engineer and Society	Apply reasoning using contextual knowledge to assess the needs of society and understand the responsibilities relevant to the engineer specializing in data science.
PO7	**GA 7:** Environment and sustainability	Understand the impact of artificial intelligence as applied to the environment, and demonstrate the knowledge for sustainable development.
PO8	**GA 8:** Ethics	Adopt ethical practices as applied to the professional world to execute responsibilities and adhere to norms of the engineering profession.
PO9	**GA 9:** Individual and Team work	Execute professional functions effectively as an individual, as well as a leader or member in diverse multidisciplinary teams.
PO10	**GA 10:** Communication	Communicate effectively with the engineering community and with society in solving complex artificial intelligence problems in terms of being able to comprehend and write effective reports, make effective presentations, as well as execute and receive clear instructions.
PO11	**GA 11:** Project Management and Finance	Demonstrate an ability to lead projects and build cost models in an interdisciplinary professional setting.
PO12	**GA 12:** Lifelong Learning	Develop learning mechanisms and inculcate the ability to prepare for lifelong learning in the context of technological change.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PEO1	1	1			2	3		1			1	2
PEO2	1	1	2						1	2		1
PEO3	1		2	3	1	1				2		2
PEO4	1						1					2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Imbibe foundational knowledge of computing theory, statistics, algorithms and programming paradigms to create innovative, scientific solutions for challenges in the field of data science and artificial intelligence.
PSO2	Prepare for recent trends in the data science industry by adapting to various computing platforms, architectures and changing software paradigms.
PSO3	Evolve new design ideas for real world problems in data science using appropriate mechanisms, design patterns, modelling methods and modern machine learning and statistical tools.
PSO4	Inculcate research attitude and aptitude in students using techniques and standards as applied to data science technology migration and evolution.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3	PSO4
PEO1	1		1	
PEO2	1		2	3
PEO3	1	2		2
PEO4	2			

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty	***Science and Technology***	**Pattern (AY-2025- 2026)**	2025
Department	**Engineering Science**	**Date (w.e.f.)**	06/01/2025
Programme	*BTech* **Computer Science and Engineering**		

SEM-II

Course Type	**Course Code**	**Course Name**	**Teaching Scheme (Hours/Week)**			**Credit**	**Marks**
			L	**T**	**P**	**C**	**Total**
BSC	BTECCSE25201	Linear Algebra and Statistics	3	1	0	4	100
ESC	BTECCSE25202	Digital Logic Design	3	0	2	4	100
ESC	BTECCSE25203	Python for Logic Development and Programming	1	0	2	2	100
VSEC	BTECCSE25204	Design Thinking and Sustainable Design	2	0	0	2	100
PCC	BTECCSE25205	Robotics and Sensor Interfacing Techniques	1	0	2	2	100
BSC	BTECCSE25206	Applied Science and Engineering Project II	0	0	4	2	100
CC	BTECCSE25207	Value Added Career Orientation II (Campus Engagement)	2	0	0	2	100
AEC	AE25CSE01A	General Proficiency and Soft Skills	1	1	0	2	100
		Total	13	02	10	20	800

Instructions, if any: 1 Theory/Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit

Head of the Department Dean

*ERP to provide document upload facility for evidencing – BoS meeting MoM, AC MoM, final approved grid with BoS chairman signature, etc.

BTECCSE25201: LINEAR ALGEBRA AND STATISTICS

Course Type	BSC	Semester	2

Teaching Scheme		Credits		Examination Scheme	
Lecture	03 Hr./Week	Lecture	03	CIE Marks	
Tutorial	01 Hr./Week	Tutorial	01	ESE Marks	
Practical/Studio	00 Hr./Week	Practical/Studio	00	Total Marks	100
Total	04 Hr./Week	Total	04		

Course Description

This course will serve as an introduction to linear algebra, statistical methods, and probability distribution. Topics covered will include Linear algebra, vector spaces, and vector field, regression, correlation, curve fitting, fundamentals of statistics, and probability distributions (discrete and continuous).

Course Outcomes

CO No.	Statement
	On completion of the course, students will be able to –
1	Develop an understanding of the algebra of matrices in order to solve applied and theoretical problems using inverses of matrices, determinants, and other algebraic operations.
2	Determine if a set of vectors is a vector space, a subspace, or a basis for a vector space, linear transformations, and understand the gradient, divergence, and curl of the vector field and compute them.
3	Apply various types of sampling methods to data collection and solve the measures of the center of data and spread data.
4	Determine an equation to represent the relationship between two (or more) linking variables and implementation in real-life situations.
5	Acquire an understanding of probability, distribution and its applications in real life problems

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	2	2	1			1						1	1			1	1,2,3
CO2	2	1							2			1	1			1	2,3,
CO3	3	1		1								1	1			1	2,3,
CO4	3	2	1	2		1			1			1	1			1	2,3
CO5	3	2	1	2		2			1			1	1			1	1,2,3

Course Content

UNIT NO. 01	MATRIX ALGEBRA	09 Hours	CO	BTL
Matrices, the rank of a matrix, System of linear equations – Consistency and Inconsistency, Gauss elimination method, Gauss Jordan method. Case studies on formation and solution of system of linear equations **Applications:** Orthogonal transformation, Eigenvalues and corresponding Eigenvectors of a matrix, Cayley- Hamilton Theorem. Role of Eigenvalues and eigenvectors in Google's page ranking			1	1,2,3
UNIT NO. 02	**VECTOR SPACES AND TRANSFORMATIONS**	**10 Hours**		
Vector Spaces: Vectors, Operations, and Properties, Definition of Vector Spaces, Subspaces, spanning set and linearly dependent and independent vectors, Basis and dimension of a vector space, Inner Product spaces, Linear transformation: Definition, Matrix of Linear Transformation, Kernel and image of a linear transformation. Applications of linear transformation in Robotics			2	2,3
UNIT NO.03	**DESCRIPTIVE STATISTICS**	**08 Hours**	3	1,2,3
Definition, Scope of Statistics, Collection and classification of data; different types of diagrams to represent statistical data; frequency distribution and related graphs and charts. **Central tendency:** Its measure and its uses. **Dispersion:** Its measure and their uses, **Moments:** Skewness and Kurtosis Statistical Analysis using Excel in case studies				
UNIT NO.04	**CORRELATION, REGRESSION & CURVE FITTING**	**08 Hours**		
Scatter diagram, Correlation coefficient with its properties and limitations; linear regression and its uses., the principle of least squares, fitting of polynomials. Analysis of real data set			4	2,3
UNIT NO.05	**PROBABILITY AND DISTRIBUTION**	**10 Hours**		
Probability: Definition, sample space, event, probability axioms, conditional probability; Bay's Theorem, Random variable. **Distributions:** Discrete Distribution-Bernoulli theorem, Binomial, and Poisson distribution. Continuous distribution-Normal distribution. Case study on Stock market analysis			5	1,2,3

Textbooks

1	Advanced Engineering Mathematics, Erwin Kreyszig, Wiley Publication, 10th Edition, 2017.
2	A First Course in Probability, Sheldon Ross, 8th Edition, 2010.
3	Higher Engineering Mathematics, B.S. Grewal, Khanna Publishers, 35th Edition, 2000.
4	Higher Engineering Mathematics, Ramana B.V., Tata McGraw Hill New Delhi, 11th Reprint, 2010.
5	Mood, A.M. Graybill, F.A. and Boes, D.C. (2007): Introduction to the Theory of Statistics, (3rd Edn.), (Reprint),Tata McGraw-Hill Pub. Co. Ltd.

Reference Books / Journal Articles / Weblink

Reference Books	
1	R.K. Jain and S.R.K. Iyenger, Advanced Engineering Mathematics, 2nd Edition, Narosa Publishing House.
2	Applied Mathematics (Vol I & II) by P. N. Wartikar& J. N. Wartikar
3	Advanced Engineering Mathematics, 7th Edition, By Peter V. O'Neil (Thomson learning publisher)
4	V. Krishnamurthy, V.P. Mainra and J.L. Arora, An introduction to Linear Algebra,Affiliated East–West press, Reprint 2005.
5	Rohatgi, V. K. & A. K. M. E Saleh (2001). Introduction to Probability and Statistics, Wiley, New York.

Sr. No	Name of Tutorial
1	Case Studies in Linear Systems: Rank, Consistency, and Orthogonality
2	From Pivot to Solution: Case Studies on linear systems and finding their solutions using Gauss and Gauss-Jordan Methods
3	The Power of Eigen: Exploring Eigenvalues, Eigenvectors with applications
4	Exploring Vector Spaces: Subspaces and the Essence of Linear Dependence in Machine Learning
5	Building Blocks of Vector Spaces: Basis, Dimension, Inner Product Spaces and their role in AI
6	Transforming Perspectives: Understanding Kernels and Images in Linear Algebra
7	Vector Fields in Focus: A Journey Through Differentiation for Feature Engineering and Transformation
8	Graph It Out: Exploring Data Analysis Through Visual Tools
9	From Data to Decisions: The Role of Mean, mode, median in Analysis of the data
10	The Spread and Shape: Exploring Dispersion and Moments in Statistics
11	Analyzing Trends: The Power of Correlation and Regression
12	From Data Points to Smooth Curves: An Introduction to Curve Fitting
13	Analyzing Random Events: A Tutorial on Binomial Distributions
14	Mastering Discrete Distributions :A Tutorial on Poisson Distributions
15	Navigating the Normal Distribution: Properties, Applications, and Insights

Unit Name	Contents	Links	CO
Matrix Algebra	Class of spaces, types of vector space, vector norms, linear product vector and theorems, and various operations that can be performed on matrices. Roles of Eigenvectors and Eigenvalues in machine learning	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013501578612809728112 94/overview	**1**
Vector spaces and transformations			**2**
Descriptive Statistics	The average of a data set and its properties; the median and mode of a data set and their properties; and the range of a data set and its properties. Then study the inter-quartile range of a data set and its properties; the variance and standard deviation of a data set and their properties	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013501585848303616114 85/overview	**3**
Correlation, Regression and curve fitting	Statistics and Probability Distribution	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013842611081396224187 90_shared/overview	**4**
Probability and Distribution			**5**

BTECCSE25202: Digital Logic Design

Course Type	ESC	Semester	2

Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hrs./Week	Lecture	3	CIE Marks	
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	
Practical/Studio	2 Hrs./Week	Practical/Studio	1	Total Marks	100
Total	5 Hrs./Week	Total	4		

Course Description

To develop a foundational understanding of number systems, logic circuits, and digital system design, including microprocessor and microcontroller fundamentals.

Course Outcomes

CO No.	At the end of this course students will be able to …..
1	Recall and represent different number systems and perform basic arithmetic operations.
2	Explain fundamental logic gates, Boolean algebra, and simplification techniques.
3	Identify ,Illustrate and Design combinational circuits
4	Identify ,Illustrate and Design sequential circuits
5	Evaluate basic architectures and functional differences of microprocessors and microcontrollers.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO3	PSO 4	
CO1	1	1										1	1			1	1,2
CO2	1	1										1				1	3
CO3	1	1	1									1				1	2
CO4	1	1	1									1	1			1	2
CO5	1	1										1				1	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.01	**Number System:**	**10 Hours**	**CO**	**BTL**
Binary numbers, Decimal numbers, hexadecimal numbers, octal numbers and number conversion, signed binary number representation: signed magnitude, 1's complement and 2's complement representation Arithmetic operations: binary addition, binary subtraction			1	2.3
Unit No.02	**Boolean Algebra for logic circuits**	**10 Hours**	2	2,3
Basic Logic variables and logic functions -NOT, AND, NOR, XOR, OR, XNOR, NAND, idealized logic gates and symbols, Truth tables, Basic theorems and properties of Boolean algebra,DeMorgan's rules, Axiomatic definition of Boolean algebra, basic theorems and properties of boolean algebra Logic minimization, representation of truth-table, don't care conditions, reduction techniques: k-maps				
Unit No.03	**Combinational Logic:**	**9 Hours**	3	3
Code Converter:- BCD, Gray code , half- adder, full adder, , comparator, Multiplexer(MUX), Demultiplexer (DEMUX), decoder				
Unit No.04	**Sequential Circuits**	**9 Hours**	4	3
Flip flops : RS, JK, D, T (truth table and excitation table) **Counters:** Ripple Binary counter, up down counter **Shift registers:** SISO, SIPO, PISO, PIPO shift registers, ring counter, Applications.				
Unit No.05	**Introduction to Microprocessor**	**7 Hours**	5	3
Introduction to von Neumann machine and Harvard architecture, RISC, CISC, Memory System (Hierarchy), Comparison of Microprocessor and Microcontroller.				

Textbooks

1. "Modern Digital Electronics", R. P. Jain, Tata McGraw-Hill, 3rd Edition
2. "Fundamentals of Digital Logic with VHDL Design", Stephen Brown, Zvonko Vranesic McGraw-Hill 2nd edition
3. Floyd, "Digital Fundamentals", Pearson Education
4. M.Morris Mano,"Digital Logic and computer design",prentice Hall of India.

Reference Books / Journal Articles / Weblink

Reference Books

1. Anand Kumar, "Fundamental of digital circuits", prentice Hall of India.
2. The Intel Microprocessors: 8086/8088, 80186, 80286, 80386 & 80486, Bary B. Brey, Prentice Hall, India 1996.

List of Experiments:

1.	To study and verify the truth table of Logic Gates (AND,OR,NOT,NAND,NOR,EXOR)
2.	Study and implement basic gates from Universal gates
3.	Verify DeMorgan's Theorems using logic circuits
4.	Minimise a given Boolean expression using K-Map and implement it using gates
5.	Design a Half Adder and Full Adder using basic gates
6.	(a) Implementation of 4:1 Multiplexer and test with input combinations (b) Design and simulate 1:4 Demultiplexer
7.	Design and verify 3-to-8 Decoder circuit
8.	Design and simulate a 1 bit Comparator
9	Simulation of RS and JK Flip-Flop and verification of truth table
10	Design of Up/Down Counter

Springboard Courses:

Course Name	Contents	Links	CO
Digital Electronics	Digital Electronics is one of the branches of Electronics. It deals with digital format of data and codes. In this, only two conditions are possible, 0 known as low logic and 1 known as high logic. Digital Electronics may also refer to the portion of an electronic circuitry responsible for digital signal conversion. Some devices in this field include multiplexers, gates, and decoders.	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138432290805350403383 7_shared/overview**	**1,2,3,4**

Electronic Circuits	Electronic Circuits and Number System is an online course where you will be learning about decimal and binary representation, Octal & hexadecimal number representation & number conversation method, Conversion Decimal-octal &hexadecimal, binary-decimal & hexadecimal, Conversion of binanry-Octal,Hexa-Binary,hex-dec,hex-octal,Oct-decimal,Octal-bina, Working of basic gates and universal gates and Applications of universal gates.	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138417572110336002961_shared/overview**	**1,2**
Sequential digital circuits	In this course it explains the sequential digital circuits. It also explains Type of Flip-Flops how to work with asynchronous input & Conversion of flip flops, shift registers, counters and much more explained in this course.	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138417768254341123873_shared/overview**	**5**

BTECCSE25203:: Python for Logic Development and Programming

Course Type-	LC	Semester	III

Teaching Scheme		Credits		Examination Scheme	
Lecture	1Hr./Week	Lecture	1	CIE Marks	
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	100
Total	3 Hr./Week	Total	2		

Course Description

On completion of the course, student will be able to –	
1	To implement the concepts of sorting and searching
2	To understand basic concepts in python
3	To develop application employing different object-oriented concepts
4	To implement linear algebra functions for arrays using NumPy
5	To demonstrate high-performance data manipulation and analysis using Pandas Library and to plot data visualization using Matplotlib and seaborn libraries

Course Outcomes

1	Analyze various sorting techniques to find concise solutions
2	Analyze various searching techniques to find concise solutions
3	To learn fundamental concepts of Python Programming and object-oriented features of Python
4	To perform data processing using NumPy library in Python
5	To analyze data using Pandas and data visualization libraries in Python Programming

Mapping of COs to POs and PSOs

CO No	POs												PSO				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1			3										1				1
CO2					3												3
CO3						2							1				3
CO4							2										3
CO5														3			3

Course Content

Unit No-1	**Fundamental Concepts in Python**	**3**
Introduction to Python, Variables, Keywords, Literals, Data types in python: List, Tuples, Dictionary, Set Operators in python, Input and Output, Control statement, Loops, Functions, Modules, File I/O, Exception, Introduction to OOP, Classes, Objects, Interfaces, Constructor, Inheritance		
Unit No-2	**Data Processing Library in Python**	**3**
NumPy, why do we need NumPy-Arrays, Shaping and transposition, Mathematical Operations, Indexing and slicing, Broadcasting, NumPy Array Attributes, Descriptive Statistics methods, matrices		
Unit No-3	**Sorting Techniques**	**3**
Time and Space complexity w.r.t. sorting algorithms such as: Selection Sort, Bubble Sort, Insertion Sort, Merge Sort, Quick Sort, and Radix Sort.		
Unit No-4	**Searching Techniques**	**2**
Searching: Linear Search, Binary Search. Fibonacci Search, Interpolation Search, Exponential Search, Ternary Search.		
Unit No-5	**Data Analysis using Python Library**	**4**
Data Analysis: Overview of Python Libraries for Data Scientists, reading data using pandas, Data frame, Data Frame data types, Data Frames attributes, Exploring data frames, Data Frames methods, group by method, filtering, Slicing, sorting, missing value, Aggregation Functions in Pandas. **Data Visualization**: Plotting libraries in python, Matplotlib: Scatter Plot, Histogram, Bar Plot. Seaborn: Scatter Plot, Histogram, Bar Plot, Box and whiskers plot, Pairwise Plots		

Textbooks

1	*Timothy Budd, "Exploring Python", Mc Graw Hill Publication, ISBN: 9780073523378, August 2010.*
2	*Peter Norton, Alex Samuel, David Aitel , etc. , "Beginning Python" , Wiley Publishing, Inc., ISBN: 978-0-7645-9654-4, August 2005.*
3	*Core Python Programming, Dr. R. Nageswara Rao, Dreamtech Press.*
4	*Beginning Python: Using Python 2.6 and Python 3.1. James Payne, Wrox publication.*
5	*"Data Structures using C",* Reema Thareja, Oxford University Press, 2nd Edition, ISBN: 9780198099307, 2015

Reference Books / Journal Articles / Weblink

Reference Books	1. *Perl Black Book, 2nd Edition: Steven Holzner, Dreamtech Press* 2. *Dana Moore, Raymond Budd, William Wright, "Professional Python Frameworks:* 3. *Web 2.0 Programming", Wrox Publication, ISBN: 978-0-470-13809-0, October 2007.*
Web resources	http://www.python.org/ www.codecademy.com/en/tracks/python https://github.com/python

List of Experiments

1	Write a Python program as basic calculator to: add, subtract, multiply, and divide two numbers (of any type). Make use of special arithmetic operators: modulus and power.
2	Write a Python program to find sum of array elements.
3	Write a Python program to swap two elements in a list.
4	Write a Python program for string manipulation operations.
5	Write a Python program to add, multiply two matrices.
6	Write a Python program to implement various sorting techniques.
7	Write a Python program to implement various searching techniques.
8	Write a Python program for merging two Dictionaries
9	Write a Python program to find maximum and minimum K elements in Tuple.
10	Write a Python program to get number of characters, words, spaces and lines in a file.

11	Write a Python program to create Employee class and its objects and display data.
12	Write a Python program to make use of all Pandas library operations.
13	Write a Python program to show all operations using NumPy Library.
14	Write a Python program to draw different plots using Matplotlib and seaborn.

Springboard Courses:

Course Name	Contents	Links	CO
Data Structures and Algorithms in Python: Sorting Algorithms	Sorting algorithms, their performance, bubble sort, selection sort, insertion sort, merge sort, quick sort	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0135015631086632969071/overview	**1**
Basics of Python	Basics of Python- Datatypes, variables, operators, selection construct, iteration construct	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0132045623015915521 12_shared/overview	**3**
Functions in Python	Define, call and utilize functions, function arguments, return values, scope and best practices for creating clean fucntions	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0143513977464913921 5/overview	**3**
Data Visualization with Python Libraries	Anaconda Navigator, Set up and Installation, NumPy in Jupiter Notebook, Array Function, NumPy Indexing and Selection, Filter Function, Python Libraries for Visualization, Matplotlib Library and its users, Plotting of Data	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138421339317698561336 5_shared/overview	**4,5**

BTECCSE25204: Design Thinking and Sustainable Design

Course Type	BSC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	02		

Course Description

This course is focusing on understanding the core principles and mindset that drive design thinking. This course will be used to complete the design thinking process to tackle real-world challenges and discover its business applications by creative and critical thinking.

Course Outcomes

CO No.	Statement
1	Explain the basic principles and mindset of Design Thinking as a human-centered approach to innovation.
2	Demonstrate the use of creative and critical thinking techniques to generate and evaluate innovative ideas.
3	Assess and improve personal creative and analytical thinking skills through structured exercises and reflection.
4	Apply the full Design Thinking process to real-world challenges, from empathy to testing and refinement, and identify relevant business use cases.
5	Collaborate and effectively use the Design Thinking framework to solve problems in team-based settings.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1	1	1	3			1	1	1	2	1	1	1					1,2,3
CO2						2	2	1	1	2	1	1					2,3
CO3							3			1		2					1,2,3
CO4									3								1,3
CO5							2	1	2		2						1,2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

COURSE CONTENT

UNIT NO. 01	Principles of Design Thinking	06 Hours	CO	BTL
What is Design Thinking? History and evolution of the concept Design Thinking vs traditional problem solving The Design Thinking mindset (empathy, experimentation, iteration) Key benefits and use cases			1	1,2,3
UNIT NO.02	**Creative & Critical Thinking Processes**	**06 Hours**		
Definitions: Creative vs Critical Thinking Left brain vs right brain thinking Thinking models: SCAMPER, Six Thinking Hats, Divergent/Convergent thinking Critical analysis, reasoning, questioning assumptions including 5 - Whys			2	2,3
UNIT NO.03	**Practicing Creative & Critical Thinking**	**06 Hours**		
Barriers to creativity and how to overcome them Personal thinking styles and creativity assessments Enhancing observation, curiosity, reframing and questioning Balancing analysis and imagination Team-based problem solving – collaboration, and psychological safety			3	1,2,3
UNIT NO.04	**The Design Thinking Framework for Collaboration**	**06 Hours**		
The 7 stages: Discover, Empathize, Define, Ideate, Prototype, Evaluate/Test, Refine Importance of cross-functional collaboration Team roles and dynamics in innovation Practical frameworks (Double Diamond, IDEO, Stanford d.school)			4	1,3
UNIT NO.05	**Applying Design Thinking to Real-World Challenges**	**06 Hours**		
Discover and empathize to gain a deep understanding of end-user needs Prototyping and low-fidelity mock-ups Testing with users and gathering feedback Iterating based on insights Design Thinking in business: product, service, process innovation Design Thinking and Sustainability Resources for ongoing learning			5	1,2,3

Textbooks

1	

Reference Books / Journal Articles / Weblink

Reference Books	
1	Design Thinking Toolkits (free) but not restricted to: Miro, Figma, Trello
2	Readings: *The Design of Everyday Things*, *Creative Confidence*, d.school guides
3	Free MOOCs or videos for continued learning – e.g. “Design Thinking and Innovation” by B K Chakravarthy (but not restricted to)

Springboard Courses:

Course Name	Contents	Links	CO
Design Thinking	This course covers the Design Thinking methodology. It highlights the importance of Design Thinking in today's business world. You will learn the Design Thinking constructs through examples and practice.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012785877065097216156 7_shared/overview	**1**
Design Thinking | Design with ChatGPT	Design with ChatGPT" addresses the evolving landscape of technology and design. ChatGPT, with its advanced natural language processing capabilities, opens new avenues for designers to explore—from automating parts of the design process to enhancing user research, data analysis and generating creative content	https://infyspringboard.onwingspan.com/web/en/app/search/learning?lang=en&q=Design%20thinking%20and%20sustainable%20design&p=0&f=%7B%7D	**2**
Sustainable Design Practices in UI	In this session, you'll explore how to integrate environmentally conscious and socially responsible principles into User Interface (UI) design. You'll learn about designing for energy efficiency, reduced digital footprint, ethical considerations, and creating long-lasting, accessible user experiences.	https://infyspringboard.onwingspan.com/web/en/app/search/learning?lang=en&q=Design%20thinking%20and%20sustainable%20design&p=1&f=%7B%7D	**3**
An Introduction to Design Thinking	In this class "An Introduction to Design Thinking", you will learn:The 5-step approach to problem solving originally developed at Stanford's Hasso Plattner Institute of Design Strategies for using the five steps (Empathize, Define, Ideate, Prototype, and Test)The three major design constraints--Desirability, Feasibility and Viability--and how they interact.	https://infyspringboard.onwingspan.com/web/en/app/search/learning?lang=en&q=design%20thinking%20%20and%20sustainable%20design&p=0&f=%7B%7D	**4**
Prototyping Design Thinking	Discover how to model prototypes in order to demonstrate the capabilities of systems and solutions.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013501551442083840712 5/overview	**5**

BTECCSE25205: Robotics and Sensor Interfacing Techniques

Course Type	PCC	Semester	2

Teaching Scheme		Credits		Examination Scheme	
Lecture	1Hrs./Week	Lecture	1	CIE Marks	
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	
Practical/Studio	2 Hrs./Week	Practical/Studio	1	Total Marks	100
Total	3 Hrs./Week	Total	2		

Course Outcomes

CO No.	At the end of this course students will be able to …..
1	Illustrate, explain and demonstrate working of elements in instrumentation system.
2	Understand and use open source resources of Arduino Development Boards and IDE
3	Develop codes to perform various operations using open source microcontroller development boards.
4	Interface development boards and develop decision making techniques

Mapping of COs to POs and PSOs

CO Number	POs												PSOs			BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	
CO1	1	1	1								1	1				2
CO2	1										1	1				2
CO3	1	1	1								1	1				3
CO4	1	1	1								1	1				2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No.01	Introduction to Embedded Systems and Microcontrollers	4 Hours	CO	BTL
Introduction to Embedded Systems and Microcontrollers, Architecture of basic microcontroller (e.g., ATmega328P), Basics of digital input/output operations, Overview of development tools: compiler, debugger, and simulator			1	2

Unit No.02	**Introduction to Arduino Board**	**3 Hours**		
Arduino: Birth, Open Source community, Functional Block Diagram, Functions of each Pin, Arduino Development Boards: IDE, Installation of Arduino IDE.			2	2
Unit No.03	**Arduino: Programming**	**5 Hours**		
Program structure: Data types, Variables and Constants, Operators, Control statements, I/O Functions, Looping Techniques, Decision Making Techniques, Designing of 1st sketch, Programming of an Arduino (Arduino ISP).			3	3
Unit No.04	**Arduino: Interfacing with devices**	**3 Hours**		
Interfacing: input- button, keypad, Serial communication, Digital and Analog input /output, Interfacing LED, Switch, 7seg LED, different sensors and output devices with Arduino.			4	2

Textbooks

1	H.S. Kalasi "Electronic Instrumentation', Tata McGraw Hill
2	Beginning Arduino Programming, Brian Evans ,Technology in Action
3	Exploring Arduino, Jeremy Blum, Wiley

Reference Books / Journal Articles / Weblink

Reference Books	
1	"Programming Arduino: Getting Started with Sketches" by Simon Monk
2	"Getting Started with Arduino" by Massimo Banzi and Michael Shiloh

LIST OF LIST OF EXPERIMENTS:
Catagory-I: Basic Interfacing
1. Introduction to Arduino Board and Arduino IDE (Installation and Setup)
2. Write a Program to Blink LED (Turn an LED on and off).
3. Write a Program to demonstrate the use of analog output to fade an LED.
4. Write a Program to read an analog input and prints the voltage to the serial monitor.
Catagory-II: Interfacing with InPut devices and sensors
5. Write a Program to Control an LED using Button.
6. Write a Program to Control Electronic Appliances using RELAY SHIELD Sensor.

7. Write a Program to measure Temperature and Humidity using DHT11 Sensor.
8. Write a program to detect motion using Motion Sensor (PIR sensor).
Catagory-III: Interfacing with Output devices and actuators
9. Write a Program to control DC motor.
10. Write a program to interface Keypad.
Catagory-IV: Interfacing with communication based devices
11. Arduino interfacing with bluetooth module
12. Arduino interfacing with wifi module.

Springboard Courses:

Course Name	**Contents**	**Links**	**CO**
Introduction to Microcontrollers	Basics of microcontrollers, architecture, I/O, etc	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01406622073202278439/overview	CO1
Getting Started With Arduino UNO	Introduction to Arduino: Arduino UNO board, its components, and how it can be used to build a wide range of projects. Setting up the environment: Programming fundamentals: Arduino programming language. variables, data types, control structures, functions, and how to interact with digital and analog pins on the Arduino UNO board. Hands-on projects: Sensor integration: temperature sensors, light sensors, and motion sensors, and learn how to interface them with the Arduino	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01380987573801779217/overview	CO2, CO3, CO4

	UNO board, data acquisition and processing. Communication and connectivity: communication protocols, such as Serial, I2C, and SPI. You'll learn how to communicate with other devices, such as displays, using these protocols. Troubleshooting and debugging: troubleshooting and debugging techniques to identify and resolve issues that may arise during the development process. Project customization and expansion:		

BTECCSE25206: Applied Science and Engineering Project II

Course Type	BSC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	04		

Course Description

This course is focusing on understanding the core principles and mindset that drive design thinking. This course will be used to complete the design thinking process to tackle real-world challenges and discover its business applications by creative and critical thinking.

Course Outcomes

CO No.	Statement
1	CO1: Execute structured testing and data collection procedures to evaluate the performance, usability, and reliability of developed prototypes or solutions.
2	CO2: Analyse and interpret quantitative and qualitative data using appropriate metrics, visualization tools, and statistical methods to identify areas for refinement.
3	CO3: Communicate project outcomes effectively through technical reports, presentations, and reflective documentation, while exploring the broader impact and career relevance of the work (e.g., entrepreneurship, patents, real-world application).

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1	1	1	3			1	1	1	2	1	1	1					1,2,3
CO2						2	2	1	1	2	1	1					2,3
CO3							3			1		2					1,2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom’s Taxonomy Level

COURSE CONTENT

	ASEP-II	30 Hours	CO	BTL
1.	Development of prototype/solution (continue from Sem I)		CO1 - 1 to 4	1,2,3
2.	Run full tests under varied conditions			
3.	Collect structured data			
4.	Identify bugs, performance issues, and usability concerns			
5.	Define evaluation metrics (e.g., speed, accuracy, energy, cost, usability)			
6.	Conduct controlled tests or user trials		CO2 – 5 to 10	
7.	Analyse data using statistical or comparative methods			
8.	Visualize results (charts, tables, dashboards)			
9.	Refine or optimize based on performance gaps			
10.	Structure and write the final technical/project report			
11.	Create a presentation deck or scientific poster			
12.	Prepare for demo day or symposium		CO3- 11 to 15	
13.	Peer and faculty evaluation			
14.	Reflective journaling on technical learning, challenges, and soft skills			
15.	Career relevance discussion: entrepreneurship, patents, real-world deployment			

Textbooks

1	
2	
3	

Reference Books / Journal Articles / Weblink

Reference Books	
1	https://www.linkedin.com/posts/muhammad-muneeb1_phd-researchpaper-activity-7341801910825287682NwS8?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA https://www.linkedin.com/posts/baak_researchwriting-abstractwriting-academicsuccess-activity-7341332655910576128 4HrH?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA https://www.linkedin.com/posts/dr-faheem-ullah_review-your-manuscript-ugcPost-7338539574849069056XHbC?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA https://www.linkedin.com/posts/samina-amin-0b9b46250_phd-research-publications-activity-7337788130202214401ovC0?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA

BTECCSE25207: Value Added Career Orientation II

Course Type	CC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	02		

Course Description:

To empower F.Y.B.Tech students with career readiness, soft skills, and holistic personal development through collaborative exploration of resources, group projects, and team-based creative assessments.

Course Outcomes:

Upon completion of this course, the student will be able to:

CO No.	Statement
1	Apply self-awareness to real-life situations such as interviews, teamwork, and presentations
2	Build a personalized system for continuous career and skill development
3	Understand and apply networking and mentoring practices
4	Practice professional communication and idea presentation in team settings
5	Demonstrate team collaboration in ideating or solving real-world scenarios

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1								1	1			3		1		1	3
CO2					1	1					2				1	1	4
CO3							1	1	1		1			1			3
CO4						3	2			2			1		1		4
CO5							1	1		3	2				1	1	5

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

COURSE CONTENT

UNIT NO. 01	DREAM RESUME – PLANNING BACKWARD FROM YOUR IDEAL ROLE	06 Hours	CO	BTL
What is a 'Dream Resume'? Role models and benchmarking; Desk research- roles, skills, qualifications, experience; Getting feedback from a faculty mentor or senior; Resume language and structure (skills-first, impact-oriented)			1	1, 2
UNIT NO.02	**CAREER ORIENTATION & FUTURE OF WORK**	**06 Hours**		
What is a learning ecosystem? Formal + informal learning; Self-paced learning tools (MOOCs, certifications, micro-skills); Setting short-term and long-term learning goals; Productivity and growth routines (journaling, habits, feedback loops)			2	2, 3
UNIT NO.03	**NETWORKS & RELATIONSHIPS – GROWING WITH PEOPLE**	**06 Hours**		
Understanding personal and professional networks; Basics of LinkedIn and digital networking; Types of mentors- peer, academic, industry; Communication do's and don'ts (online + offline)			3	3, 4
UNIT NO.04	**IDEA TO ACTION – SMALL PROJECTS, BIG LEARNING**	**06 Hours**		
Ideation techniques- brainstorming, problem framing; Understanding small problems with real impact; Presenting an idea simply and clearly (1-minute pitch); Taking feedback and refining solutions			4	2, 3
UNIT NO.05	**MY CAREER TOOLKIT – TYING IT ALL TOGETHER**	**06 Hours**		
Creating your career asset kit- Resume + LinkedIn + Learning Plan; Reviewing career goals from Semester 1 and refining them; Final reflections- who I am, what I want, where I'm going; Peer and mentor feedback loop			5	4, 5

Reference Books:

1.	"What Color Is Your Parachute? for College: Pave Your Path from Major to Meaningful Work" by Katherine Brooks, Ten Speed Press (2021).
2.	"The Start-up of You: Adapt, Take Risks, Grow Your Network" by Reid Hoffman and Ben Casnocha, Random House Business (2013).
3.	"Creative Confidence: Unleashing the Creative Potential Within Us All" by Tom Kelley and David Kelley (2013).

Infosys Springboard Courses:

Unit	Topics	Springboard Course	CO
DREAM RESUME – PLANNING BACKWARD FROM YOUR IDEAL ROLE	Self Management	Management of Self https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013301009606205440430/overview	1
CAREER ORIENTATION & FUTURE OF WORK	Personal branding and time management	Introduction to personal branding https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0142227405746012161608/overview	2
NETWORKS & RELATIONSHIPS – GROWING WITH PEOPLE	Professional Etiquettes and workplace communication	Professional Étiquettes https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01384266175560908819951_shared/overview	3
IDEA TO ACTION – SMALL PROJECTS, BIG LEARNING	Entrepreneurship Essentials	The Essential Guide to Entrepreneurship by Guy Kawasaki https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_014157720196939776401/overview	4
MY CAREER TOOLKIT – TYING IT ALL TOGETHER	Career planning and interview preparation	Career Planning and Interview Preparation https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01419086838180249673/overview	5

AE25CSE01A: General Proficiency and Soft Skills

Course Type	AEC	Semester	

Teaching Scheme		Credits		Examination Scheme	
Lecture	1 Hr./Week	Lecture	1	CIE Marks	
Tutorial	1 Hr./Week	Tutorial	1	ESE Marks	
Practical/Studio	0 Hr/ Week	Practical/Studio	0	Total Marks	100
Total	2 Hr./Week	Total	2		

Course Description

This rigorous course targets to imbibe basic and advanced English Language skills among undergraduate students. Students will get a robust understanding of various essential concepts in English Language Usage from a practical point of view. This course will help to bring crucial professional development in the behaviour of the students, which will enable them to cope with social, formal and organizational situations.

Course Outcomes

CO No.	Statement
1	Students will demonstrate foundational knowledge of effective communication principles, including an understanding of verbal and non-verbal communication, pronunciation, and the importance of active listening.
2	Students will be able to interpret and summarize various types of written content, showcasing the ability to identify main ideas, supporting details, and employing effective reading comprehension strategies.
3	Students will apply their knowledge in practical scenarios, including crafting well-structured emails, engaging in effective conversations, and producing creative writing pieces that demonstrate a command of grammar and punctuation rules.
4	Students will analyze and evaluate communication scenarios in professional settings, such as job interviews and online communication platforms, identifying effective practices and recognizing the impact of language choices on communication outcomes.
5	Through a final project, students will synthesize their acquired language skills by creating an integrated and practical output. This may involve developing a comprehensive communication portfolio or executing a real-world project that requires the application of various language elements.

Mapping of COs to POs and PSOs

CO Number	POs												BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11	PO 12	
CO1	1	-	-	-	-	-	-	-	1	2	-	1	L1, L2
CO2	1	-	-	-	-	-	-	1	-	3	-	1	L2

CO3	-	-	1	-	-	-	-	2	1	2	-	2	L3
CO4	-	-	-	1	-	1	2	1	1	3	-	1	L3, L4, L5
CO5	-	-	-	1	-	1	1	1	1	3	1	1	L5, L6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Corporate Communication	**Hours 5**	**CO**	**BTL**
1.1 Basic concepts of Corporate Communication ● Definition and importance of Corporate Communication ● Overview of organizational channels of communication ● Verbal and non- verbal communication ● Barriers to communication and strategies to overcome.: Case studies (Principles of Effective Communication) ● Principles of effective communication 1.2 Speaking Skills: ● How to introduce yourself and others? - Personal Information, achievements, family background Soft Skills: 1.3 Self-Evaluation for Career Planning: Short term, Long term goals. Setting SMART goals, process of performing SWOT Analysis 1. 4. Interpersonal Skill Development: Positive Relationship, Positive Attitudes, Handling Situations 1.5. Professional etiquettes: meaning and significance, organizational ethics, Telephonic etiquettes, professional etiquettes, grooming.			1,3	1,3
Unit No 2	Writing Skills for Corporate Use	**Hours 5**	**3**	**3**
2.1 Email and Business Writing ● Formal and informal email writing ● Professional tone and language ● Common business writing practices 2.2 Application Letter writing 2.3 Resume Writing 2.4 Creative writing				
Unit No 3	Corporate Register	**Hours 5**	3,4	3,5,6
3.1 Comprehensive reading: Skimming, scanning, intensive and extensive 3.2 Listening skills: Process of listening, Types, Barriers in listening, Tips for being a good listener 3.3 Team building: process of team building, significance and methods to develop				

Unit No 4	ESS: English for Specific Situations	**Hours 5**	**2**	**4**
Corporate vocabulary in various contexts, Corporate jargons, morphology, syntax, Synonyms and antonyms, affixation, Idiom and phrases				
Unit No 5	**Presentation skills**	**Hours 5**	**4,5**	**5**
● Flow chart, poster presentation, role plays, mime, meetings summary presentation ● Group Discussion: Leadership, Persuasive talk, statement making ● Debate, extempore, ● Do's and don't of a Presentation ● Public speaking: do's and don'ts in public speech				

Textbooks

1	*The Handbook of Communication Skills edited by Owen Hargie*
2	Business English by N Gupta et al 2003

Reference Books / Journal Articles / Weblink

1	Business English, Pearson, 2008.
2	*English and Soft Skills BY S P Dhanavel*
3	*Cambridge Soft Skills and Employability Skills by Sabina Pillai, Agna Fernandez*
4	Entry from Backside Only by Binoo K John
5	University Grammar of English by
6	English Phonetics for Indian Students by Balsubramanian

Guidelines for Laboratory/Studio Conduction *<If Applicable>*

<Instructions>

List of Experiments/Studio Assignments, etc

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	Journal Submission		
2	MCQ		
3	Blog Writing		
4	Presentation		
5	Resume and cover letter		
6	Poster making		

Springboard Courses:

Course Name	Contents	Links	CO
English Communication	English Communication	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01419631328410828834/overview	1,2,3,4
Barriers To Communication	Types of barriers to Communication	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138478099313950725430 4_shared/overview	1,4
Basics of Business Communication	Importance and Basic Business Communication	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01268876808363212830 8_shared/overview	1,2,3,4,5
Take Your English Communication Skills to the Next Level	Types and Practice of communication Skills	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_014235412442923008316 7/overview	1,2,3,4,5
Communication skills for engineering scholars	The course teaches engineering scholars how to improve their communication skills, based on theory, examples and practical exercises. Topics include basics of rhetoric (how to be clear), basics of argumentation theory (how to be persuasive), public speaking, grant writing, scientific writing. The main target are engineering scholars; the first 3 weeks, though, can be useful to anyone wishing to improve their communication skills.	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_014235408148078592297 8/overview	2,3,4